

The designer has worked smartly to minimise memory usage to as minimum as possible, intended for MCUs. Minimal random access memory (RAM) usage, minimal code and quick execution, in this order, are the driving factors behind the design of MicroPython. A simple and fast mark-sweep garbage collector manages memory, taking less than four minutes to perform a full collection.

Objects are stored efficiently and many functions can be written to use no heap memory, thus requiring no garbage collection either. This is well-illustrated in the case of integers that fit in 31 bits, which do not allocate an object on the heap and require memory only on the stack. Even exceptions are handled with minimum memory using a variant of `setjmp/longjmp` instructions. A cut-down build of MicroPython can work with around 75kB of read-only

Investing in innovation

Ventures based on MicroPython are gaining ground. Below is a look at three projects on Kickstarter that have been established over time:

Pyboard. This is an electronics development board that runs MicroPython and is based on STM32F405 MCU. The board is available through MicroPython website (<http://micropython.org/>), Adafruit, MC Hobby and Robot Domestic.

Machine vision with Python. A camera board on the face of it, OpenMV Cam is a complete vision solution that uses a simple integrated development environment, rapidly integrates into any hardware project and supports shields for expanded capabilities.

WiPy. It is a killer combination of MicroPython software and robust, low-power Wi-Fi hardware in one tiny, self-contained package. It has been extended to Pycom, a company started by Daniel Campora, and aims at making communication-related boards that run MicroPython. Plans are also on for another board, LoPy, which has LoRa capabilities along with Wi-Fi and Bluetooth.

memory (ROM) and 8kB of RAM.

Putting it to maximum use

MicroPython is provided under MIT open source licence. The software package, apart from the files for use with Pyboard, includes a Unix version, one each for ARM, Teensy 3.1, 16-bit PIC and Texas Instruments'

CC3200 MCUs.

In addition to the above, there is also a minimal port that acts as a starting point for porting MicroPython to any MCU chosen by you. **EFY**

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